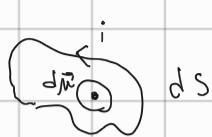


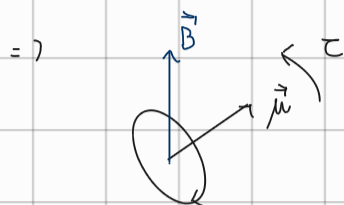
MOMENTO ANGOLARE INTRINSECO: SPIN

Princ. Equiv. AMPERE:



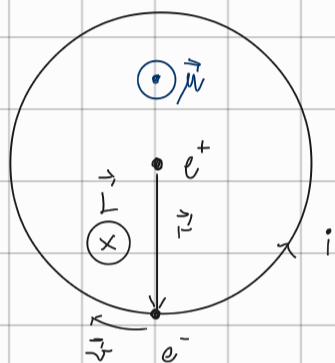
$$\Leftrightarrow \uparrow \boxed{\frac{2}{5}}$$

$$d\vec{\mu} = i ds \vec{\mu}_h$$



$$\vec{\tau} = \vec{\mu} \wedge \vec{B}, \quad U_{dip.} = -\vec{\mu} \cdot \vec{B}$$

→ FISICA ATOMICA:



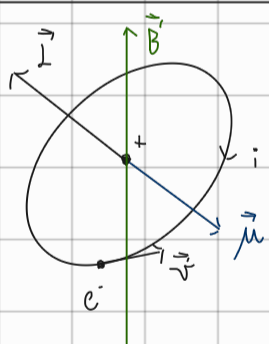
$$\vec{\mu} = i S \vec{\mu}_h$$

$$i = \frac{e}{T} = \frac{e}{\frac{2\pi r}{v}} = \frac{ev}{2\pi r}$$

$$\vec{\mu} = \frac{ev}{2\pi r} \pi r^2 \vec{\mu}_h = \frac{ev}{2} r \vec{\mu}_h =$$

$$= \frac{e}{2m} \underbrace{mvr}_{|\vec{L}|} \vec{\mu}_h = -\frac{e}{2m} \vec{L} = \vec{\mu}$$

Nella fisica delle part. $\vec{L} \Leftrightarrow \vec{\mu}$, $\gamma = -\frac{e}{2m} \Rightarrow$ RAPPORTO GIROMAGNETICO



$$\vec{\tau} = \vec{\mu} \wedge \vec{B} = -\frac{e}{2m} \vec{L} \wedge \vec{B} = \frac{e}{2m} \vec{B} \wedge \vec{L}$$

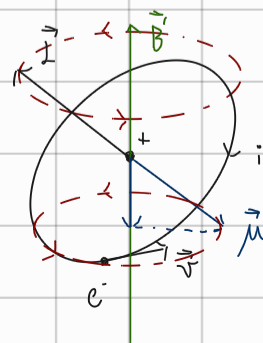
$$\frac{d\vec{L}}{dt} = \vec{\tau} \text{ (MOM. TORCENTE)} = \frac{e}{2m} \vec{B} \wedge \vec{L}$$

$$\frac{d\vec{L}}{dt} = \frac{dL}{dt} \hat{L} + \vec{\omega}_L \wedge \vec{L} = \frac{e}{2m} \vec{B} \wedge \vec{L}$$

$$a. |\vec{L}| = \text{cost}$$

$$b. \frac{d}{dt} (\vec{L} \cdot \vec{B}) = 0 \Rightarrow \angle \vec{L} \text{ e } \vec{B} \text{ RESTA COSTANTE}$$

$$c. \vec{L} \text{ RUOTA ATTORNO } \vec{B} \Rightarrow |\vec{\omega}_L| = \frac{e|\vec{B}|}{2m}$$



⇒ PRECESSIONE di LARMOR

DIAMAGNETISMO

$\omega_2 = \frac{eB}{2m} \approx 10^{11} \div 10^{12} \text{ rad/s} \ll \omega_0 = 10^{16} \text{ rad/s}$ (precessione lenta e trascurabile)

CAMPO MAG. non unif.: $\vec{F} = -\nabla U_{\text{dip}} = -\nabla (-\vec{\mu} \cdot \vec{B}) = \left\{ \vec{B} = \vec{B}(z) \vec{u}_z \right\} \Rightarrow$

$\vec{F} = \mu_z \frac{\partial B_z}{\partial z} \vec{u}_z$

UNITÀ NATURALE $\Rightarrow \hbar$ [mom. ang.]

$\vec{\mu} = -\frac{e}{2m} \vec{L} = -\frac{e\hbar}{2m} \frac{\vec{L}}{\hbar} = -g_L \mu_B \frac{\vec{L}}{\hbar}$

FATTORE g-ORBITALE $g_L = 1$

$\mu_B = 9,3 \cdot 10^{-24} \text{ J/T} = 5,8 \cdot 10^{-5} \text{ eV/T}$

$\Rightarrow L^2 = \hbar^2 l(l+1) \Rightarrow |\vec{\mu}| = \mu_B \sqrt{l(l+1)}$

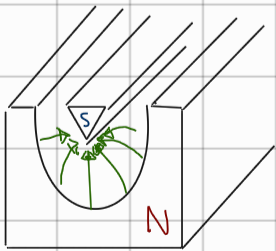
$L_z = \hbar m \Rightarrow \mu_z = -g_L \mu_B \frac{L_z}{\hbar} = -\mu_B m$

$\Rightarrow \vec{F} = \mu_z \frac{\partial B_z}{\partial z} \vec{u}_z = -\mu_B m \frac{\partial B_z}{\partial z} \vec{u}_z$

scel. campo mag. non unif.

F è in funz. di m

1923 Stern - Gerlach



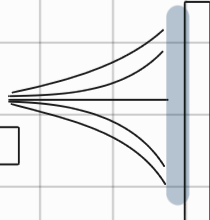
$\frac{\partial B_z}{\partial z} \approx 1000 \text{ T/m}$ (molto più forte verso la punta)

FORNO

↓
Vapori di
Atomi

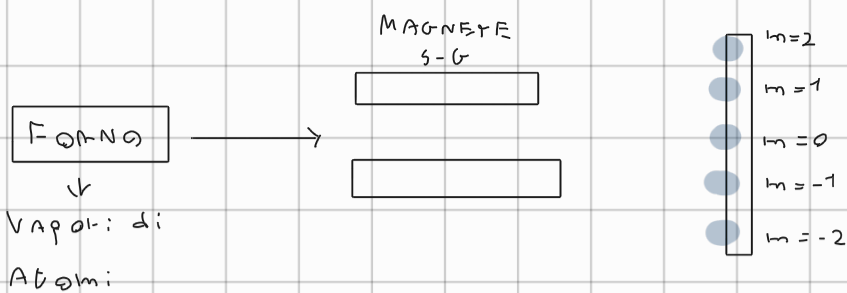
MAGNETE
S-G

Schermo

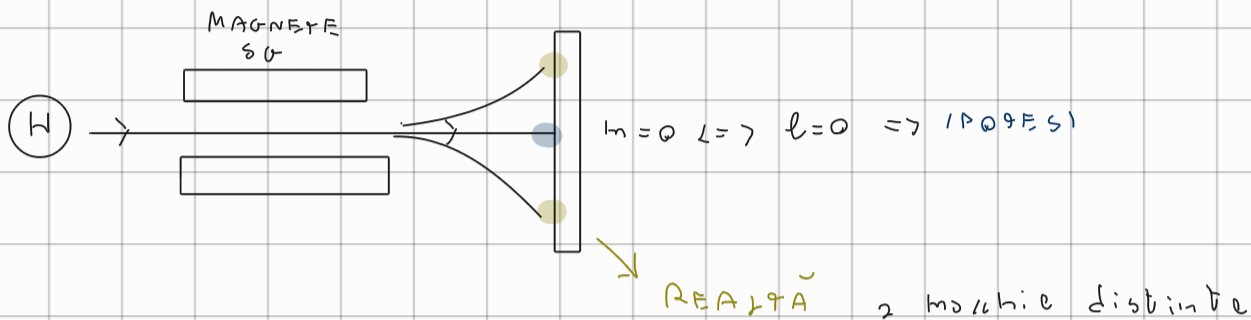


① Fisica Classica

μ_z non quantizzato



② QUANT.
 l_z QUANTIZ.
 $l=2$



1. Magnet: $l \neq 0$ Ma dovrai avere num. di macchie dispari. X

2. Magnet: dovuto alle cariche nel nucleo

$$|\vec{\mu}_B| \propto \mu_B \Rightarrow \mu_B^N = \frac{e\hbar}{2m_p} \approx 10^{-3} \text{ volte più piccolo di } \mu_B^e \quad \text{X}$$

$$\Rightarrow F_z = \mu_z \frac{\partial B_z}{\partial z}$$

MOMENTO DI DIP. MAGN. di SPIN $\rightarrow$ Mom. mag. intrinseco: SPIN

1) S-G non si può fare solo con $e^- \rightarrow \vec{F}_L = g \vec{v} \wedge \vec{B}$

2) S-G è stato fatto con atomi di Ag $\rightarrow$ c'è solo rel. SS $\Rightarrow l=0$

1925 $\rightarrow$ Cos'è lo SPIN?



$\Rightarrow \sqrt{2} \gg 1$ X

SPIN è un grado di lib. quantistico che non ha analogo classico

$\vec{F}_z = \mu_B \frac{\partial B_z}{\partial z}$ (Noto (progettato) o. l. magnetico)
 $\Rightarrow$ Risultato $\Rightarrow \mu_z = \pm \mu_B$ $\rightarrow$ Mom. di dip. mag. intrinseco quantizzato per e^-
 Misura la deflessione

$\vec{\mu}_e = -g_e \mu_B \frac{\vec{L}}{\hbar} \Leftrightarrow \vec{\mu}_s = -g_s \mu_B \frac{\vec{S}}{\hbar}$ (introduco per analogia)

$(\mu_L)_z = -g_L \mu_B m_L \Leftrightarrow (\mu_S)_z = -g_S \mu_B m_S$

$\hat{S} \Leftrightarrow \hat{L}$
 $\rightarrow$ Op. di SPIN

(S: può dim. che) QUALUNQUE MOM. ANG. $\vec{J}$:

$\Rightarrow [\hat{J}^2, \hat{J}_i] = 0 \quad e \quad [\hat{J}_i, \hat{J}_j] = \epsilon_{ijk} \hbar \hat{J}_k$

(legge di simmetria)

$J^2 = \hbar^2 j(j+1) \Rightarrow j = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}, \dots$

$m_j = -j, -j+1, \dots, +j$

($\hat{L}$ è un sottoinsieme dei mom. $\vec{J}$ generali)

$[\hat{L}^2, \hat{L}_x] = [\hat{L}^2, \hat{L}_y] = [\hat{L}^2, \hat{L}_z] = 0 \Leftrightarrow [\hat{S}^2, \hat{S}_x] = \dots = 0$

$[\hat{L}_i, \hat{L}_j] = \epsilon_{ijk} \hbar \hat{L}_k \Leftrightarrow [\hat{S}_i, \hat{S}_j] = \epsilon_{ijk} \hbar \hat{S}_k$

$\hat{L}^2 = \hbar^2 l(l+1) \Leftrightarrow \hat{S}^2 = \hbar^2 s(s+1)$

$L_z = \hbar m_l \Leftrightarrow S_z = \hbar m_s$

$m_l = -l, \dots, +l \Leftrightarrow m_s = -s, \dots, +s$

$\Rightarrow (\mu_S)_z = \pm \mu_B = -g_S \mu_B m_S$
 Specimanb. $\left. \begin{array}{l} m_S = +\frac{1}{2} \\ m_S = -\frac{1}{2} \end{array} \right\} \Rightarrow g_S = 2$ Due macchine

Se $m_S = 1 \Rightarrow m_S = 0, -1 \Rightarrow$ n. di macchine dispari (m_S solo di 1)

$$m_s = -\frac{1}{2}, \frac{1}{2} \Rightarrow S = \frac{1}{2} \Rightarrow \text{Num. quant. di spin è semintero}$$

↳ TEOR. di SCHRÖDINGER non PREVEDE LO SPIN

1918] DIRAC Teoria RELATIVISTICA: EQ. di DIRAC $\Rightarrow$ SCH. + $E = \sqrt{p^2 c^2 + m^2 c^4} + V$

$\Rightarrow$ Si deduce naturalm. dall'eq. lo spin

SPIN $\Leftrightarrow$ RELATIVITÀ

- Elett., Prot., Neut. $\Rightarrow S = \frac{1}{2}$
- MESONI π (quark + antiquark) $= S = 0$
- Fotoni $\Rightarrow S = 1$
- BARIONI Δ (3 quark) $S = \frac{3}{2}$

$$\text{ELETTRONI: } S = \frac{1}{2}, m_s = \pm \frac{1}{2} \Rightarrow S_z = \pm \frac{\hbar}{2}$$

$$\Rightarrow \text{SPAZIO di HILBERT (2D): } \vec{w}_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \vec{w}_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\vec{w} = \alpha \vec{w}_1 + \beta \vec{w}_2, |\alpha|^2 + |\beta|^2 = 1$$

$$|w\rangle = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$$

$$\langle w| = (w_1^* \ w_2^*)$$

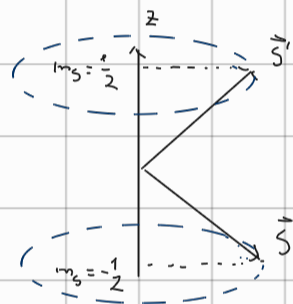
ORA GLI O.P. SONO MATRICI

$$\hat{Q} \Rightarrow \hat{Q} = \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{bmatrix}$$

$$\hat{Q}^\dagger = ? \quad \langle \hat{Q}^\dagger v | w \rangle = \langle v | \hat{Q} w \rangle$$

$$\Rightarrow \hat{S}^2, \hat{S}_x, \hat{S}_y, \hat{S}_z \Rightarrow [\hat{S}^2, \hat{S}_x] = [\hat{S}^2, \hat{S}_y] = [\hat{S}^2, \hat{S}_z] = 0$$

$$[\hat{S}_i, \hat{S}_j] = i\hbar \epsilon_{ijk} \hat{S}_k$$



SPINORI: Stato del sistema = FUNZIONE D'ONDA

$$\hat{S}_z \Rightarrow [\hat{S}^2, \hat{S}_z] = 0$$

Autovettori di: $\hat{S}_z$

$$\begin{aligned} &\rightarrow \text{SPIN-UP} \Rightarrow \chi_+ = |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow S_z = \hbar/2 \\ &\rightarrow \text{SPIN-DOWN} \Rightarrow \chi_- = |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{aligned}$$

SPINORE GENERALE: $|\chi\rangle = a|\uparrow\rangle + b|\downarrow\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$, $|a|^2 + |b|^2 = 1$

$$\rightarrow \langle \chi | = (a^* \ b^*)$$

$$\Psi = \sum_n c_n \phi_n, \quad |a|^2 = \mathcal{P}(S_z = \frac{\hbar}{2}), \quad |b|^2 = \mathcal{P}(S_z = -\frac{\hbar}{2})$$

$\nwarrow (a, b)$

$$\Rightarrow \hat{S}_z = \hbar m_s = \pm \frac{\hbar}{2} \quad \hat{S}_z |\uparrow\rangle = S_z \begin{pmatrix} 1 \\ 0 \end{pmatrix} = +\frac{\hbar}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = +\frac{\hbar}{2} |\uparrow\rangle$$

$$\hat{S}_z |\downarrow\rangle = \hat{S}_z \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{\hbar}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{\hbar}{2} |\downarrow\rangle$$

$$\Rightarrow \boxed{\hat{S}_z = \frac{\hbar}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}} \quad \Rightarrow \text{MATRICE DIAGONALE}$$

$$s = \frac{1}{2} \Rightarrow s^2 = \hbar^2 s(s+1) = \frac{3}{4} \hbar^2$$

$$\begin{cases} \hat{S}^2 |\uparrow\rangle = \hat{S}^2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{3}{4} \hbar^2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \hat{S}^2 |\downarrow\rangle = \hat{S}^2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{3}{4} \hbar^2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{cases}$$

$$\Rightarrow \boxed{\hat{S}^2 = \frac{3}{4} \hbar^2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}$$

$\Rightarrow$

$$\hat{S}_x = \frac{\hbar}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \hat{S}_y = \frac{\hbar}{2} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad (\text{2 diff. di 2 MAT. NON DIAGONALI})$$

MATRICI di PAULI:

$$\hat{\sigma}_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow \hat{S}_x = \frac{\hbar}{2} \hat{\sigma}_x, \quad \hat{\sigma}_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \Rightarrow \hat{S}_y = \frac{\hbar}{2} \hat{\sigma}_y$$

$$\hat{\sigma}_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \Rightarrow \hat{S}_z = \frac{\hbar}{2} \hat{\sigma}_z$$

(Queste matrici sono legate al modo di
ruotare dei vett. nell'alg. di Lie)

$\Rightarrow$ MAT. HERMITIANE e UNITARIE

$$\hookrightarrow (\hat{\sigma}_i^\dagger \hat{\sigma}_j = \delta_{ij})$$

$| \uparrow \rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ (AUTOST. di $\hat{S}_z$) è autost. di $\hat{S}_x$?

$$\hat{S}_x = \frac{\hbar}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \xrightarrow{\hat{S}_x} \frac{\hbar}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{\hbar}{2} | \downarrow \rangle \Rightarrow \text{NON È AUTOSTATO}$$

di $\hat{S}_x$ (non l'è di
sé stesso)

Trovo autost. di $\hat{S}_x$:

$$\det(\hat{S}_x - \lambda \mathbb{I}) = \begin{vmatrix} -\lambda & \hbar/2 \\ \hbar/2 & -\lambda \end{vmatrix} = 0 \quad \lambda^2 - \left(\frac{\hbar^2}{2}\right) = 0 \Rightarrow \boxed{\lambda = \pm \frac{\hbar}{2}} \text{ AUTOV. di } \hat{S}_x$$

$$\rightarrow \text{AUTOVET:} \quad \begin{bmatrix} -\lambda & \hbar/2 \\ \hbar/2 & -\lambda \end{bmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0 \quad \lambda = +\hbar/2$$

$$\Rightarrow \begin{bmatrix} -\hbar/2 & \hbar/2 \\ \hbar/2 & -\hbar/2 \end{bmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0 \quad -\frac{\hbar}{2} a + \frac{\hbar}{2} b = 0 \Rightarrow a = b \Rightarrow | \chi_1 \rangle = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

UN AUTOV. di $\hat{S}_x$

$$\lambda = -\hbar/2 \Rightarrow \begin{bmatrix} \hbar/2 & \hbar/2 \\ \hbar/2 & \hbar/2 \end{bmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0 \quad \hbar/2 a + \frac{\hbar}{2} b = 0 \Rightarrow a = -b \Rightarrow | \chi_2 \rangle = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$$

$$\Rightarrow [\hat{S}_z, \hat{S}_x] \neq 0$$

MISURA delle COMPONENTI dello SPIN:

$$|X\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\langle \hat{S}_z \rangle = \langle X | \hat{S}_z | X \rangle = \begin{pmatrix} a^* & b^* \end{pmatrix} \frac{\hbar}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \frac{\hbar}{2} (|a|^2 - |b|^2)$$

$$\Downarrow$$

$$\left(\int \psi^* \hat{S}_x \psi \right)$$

$$\langle \hat{S}_z \rangle = |a|^2 \frac{\hbar}{2} + |b|^2 \left(-\frac{\hbar}{2} \right)$$

$\langle A \rangle = \int \rho A_n$ Solito def. di medio di una grandezza

$$|X\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \langle \hat{S}_z \rangle = \begin{pmatrix} 1 & 0 \end{pmatrix} \frac{\hbar}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\hbar}{2} \quad (\text{come al solito } 100\%)$$

$$\rightarrow \langle \hat{S}_x \rangle = \langle X | \hat{S}_x | X \rangle = \begin{pmatrix} 1 & 0 \end{pmatrix} \frac{\hbar}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0 \Rightarrow \langle \hat{S}_x \rangle = 0$$

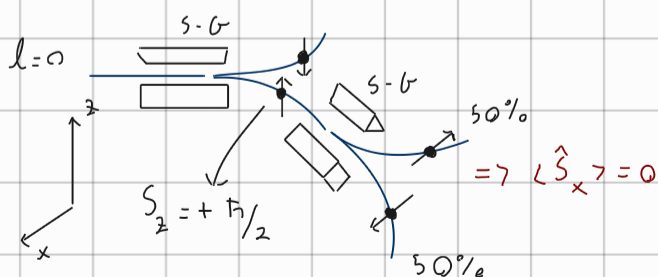
Posso scrivere $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ come sovrapp. di autost. di $\hat{S}_x \Rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} = a \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} + b \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$

$$\Rightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} \Rightarrow \text{Se sono in uno stato } \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ } 50\% \text{ misura } |\uparrow\rangle$$

$$50\% |\downarrow\rangle \Rightarrow \langle \hat{S}_x \rangle = 0$$

(Se sono in un autost. di $\hat{S}_z$ non posso conosc. con prec. S_x e S_y)

$$(\mu_s)_z = -g_s \mu_B m_s \quad m_s = -\frac{1}{2} \Rightarrow (\mu_s)_z > 0$$



$$m_s = +\frac{1}{2} \Rightarrow (\mu_s)_z < 0$$

$$F_d(\mu_s)_z$$

Il mom. di dip. mag. è invers. risp. allo spin

Se mettessi di nuovo un SG dopo la mis. di $\hat{S}_x$ troverei $\hat{S}_z$ al 50%

$$\hat{S} = \frac{\hbar}{2} \hat{\sigma} = \frac{\hbar}{2} (\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z)$$

Atomo ID:

$$w_{nlm}(\vec{r}) \rightarrow q = \{n, l, m\} \rightarrow w_q(\vec{r}) \in \mathcal{L}^2(\mathbb{R}^3)$$

$$|x\rangle = \begin{pmatrix} a \\ b \end{pmatrix} \quad a, b \in \mathbb{C} \quad , \quad |a|^2 + |b|^2 = 1 \quad \text{METTO INS.}$$

$$\Rightarrow w(\vec{r}, s_z) = \begin{pmatrix} a w_{q_1}(\vec{r}) \\ b w_{q_2}(\vec{r}) \end{pmatrix} \Rightarrow \int |a w_{q_1}(\vec{r})|^2 d\vec{r} + \int |b w_{q_2}(\vec{r})|^2 d\vec{r} = 1$$

Es:

$$a. \quad w(\vec{r}, s_z) = w_{100} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow \text{15} \quad \boxed{\uparrow}$$

$$b. \quad w(\vec{r}, s_z) = w_{100} \left[\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] \Rightarrow \text{50\%} \quad \boxed{\uparrow} \quad \text{50\%} \quad \boxed{\downarrow}$$

$$c. \quad w(\vec{r}, s_z) = \frac{1}{\sqrt{2}} w_{100} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} w_{200} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow \text{50\%} \quad \boxed{\uparrow} \quad , \quad \text{50\%} \quad \boxed{\downarrow}$$

SOMME DI MOMENTI ANGOLARI:

$$\vec{L} \Rightarrow L^2 = \hbar^2 l(l+1) \quad L_z = \hbar m_l \quad m_l = -l \dots l$$

$$\hat{S} \Rightarrow S^2 = \hbar^2 s(s+1) \quad S_z = \hbar m_s \quad m_s = \pm \frac{1}{2}$$

$$\text{GENERALICO } \vec{J} \Rightarrow [\hat{J}^2, \hat{J}_x] = [\hat{J}^2, \hat{J}_y] = [\hat{J}^2, \hat{J}_z] = 0, \quad [\hat{J}_i, \hat{J}_j] = i\hbar \epsilon_{ijk} \hat{J}_k$$

$$J^2 = \hbar^2 j(j+1) \quad J_z = \hbar m_j \quad m_j = -j \dots j$$

$$\begin{array}{ccc} \vec{J}_1 & + & \vec{J}_2 = \vec{J} \\ \downarrow & & \downarrow \end{array} \quad (\text{Per fare la somma dovrai considerare tutte le componenti})$$

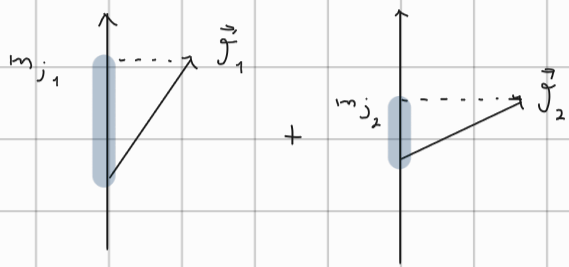
$$(j_1, m_{j_1}) \quad (j_2, m_{j_2})$$

TEOREMA di SOMMA dei MOMENTI ANGOLARI:

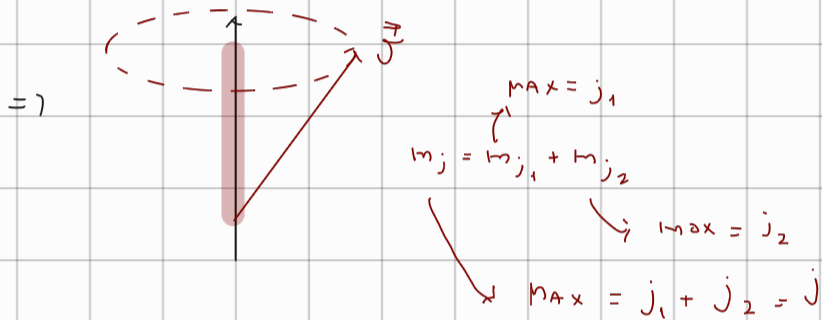
$$j \text{ (num. quanta)} = |j_1 - j_2| \dots j_1 + j_2 \quad \text{saltando di 1}$$

Es:

$$j_1 = 1 \quad j_2 = 3 \quad \Rightarrow \quad j = 2, 3, 4 \rightarrow \begin{matrix} \nearrow m_j = -2, \dots, 2 \\ \searrow m_j = -3, \dots, 3 \end{matrix}$$



Posso sommarle le proiezioni.



$$\vec{J}_1: \begin{aligned} [\hat{J}_1^2, J_{1i}] &= 0 \quad i = x, y, z \\ [\hat{J}_1, \hat{J}_{1j}] &= i \hbar \epsilon_{ijk} \hat{J}_{1k} \end{aligned}$$

$\vec{J}_1$ & $\vec{J}_2$ COMPLETELY INDEPEND.

$[\hat{J}_1^2, \hat{J}_2^2] = 0$

$$\vec{J}_2: \text{stcsho } \text{cosh}$$

$$[\hat{J}_1^2, \hat{J}_2^2] = [\hat{J}_2^2, \hat{J}_1^2] = 0$$

$$[\hat{J}_{1i}, \hat{J}_{2K}] = 0 \quad i, K = x, y, z$$

$= \gamma \hat{J}_1^2, \hat{J}_{12}, \hat{J}_2^2, \hat{J}_{23}$ COMMUTANO MUTUAMENTE (HANNO UNA BASE di AUTOVAL. in

BASE DISACCOPIATA

COMUNE)

= 7 $|j_1, m_{j_1}, j_2, m_{j_2}\rangle$ (indice 1 subastato transito: m. q. m. t.)

Q.P.

$$A \cup \emptyset = A.$$

Autov.

AUTOT.

$$\begin{aligned} \Rightarrow \hat{J}_1^2 |j_1, m_{j_1}, j_2, m_{j_2}\rangle &= \hbar^2 j_1(j_1+1) |j_1, m_{j_1}, j_2, m_{j_2}\rangle \\ \hat{J}_{1z} |j_1, m_{j_1}, j_2, m_{j_2}\rangle &= \hbar m_{j_1} |j_1, m_{j_1}, j_2, m_{j_2}\rangle \\ \hat{J}_2^2 |j_1, m_{j_1}, j_2, m_{j_2}\rangle &= \hbar^2 j_2(j_2+1) |j_1, m_{j_1}, j_2, m_{j_2}\rangle \\ \hat{J}_{2z} |j_1, m_{j_1}, j_2, m_{j_2}\rangle &= \hbar m_{j_2} |j_1, m_{j_1}, j_2, m_{j_2}\rangle \end{aligned}$$

$$\Rightarrow \hat{H} \psi_{nlm} = E_n \psi_{nlm} \Rightarrow \hat{H} |nlm\rangle = E_n |nlm\rangle$$

$$E_{5,} \psi = \frac{1}{\sqrt{2}} \psi_{100} + \frac{1}{\sqrt{2}} \psi_{210} = \frac{1}{\sqrt{2}} |100\rangle + \frac{1}{\sqrt{2}} |210\rangle$$

$$L^2 \psi_1^0 = \hbar^2 l(l+1) \psi_1^0 \Rightarrow L^2 |lm\rangle = \hbar^2 l(l+1) |lm\rangle$$

$$\hat{J}^2 = (\hat{J}_1 + \hat{J}_2)^2 = \hat{J}_1^2 + \hat{J}_2^2 + 2 \underbrace{\hat{J}_1 \hat{J}_2}_{\substack{J_{1x} J_{2x} + J_{1x} J_{2y} + J_{1y} J_{2x} + J_{1y} J_{2y} + J_{1z} J_{2z} + J_{2z} J_{1z}}}$$

$$[\hat{J}^2, \hat{J}_1^2] = [\hat{J}^2, \hat{J}_2^2] = 0$$

$$[\hat{J}_1^2, \hat{J}_2] = [\hat{J}_2^2, \hat{J}_1] = 0$$

VERIFICO DOPO

$$[\hat{J}^2, \hat{J}_{1z}] \neq 0 \quad [\hat{J}^2, \hat{J}_{2z}] \neq 0$$

$$\hat{J}^2, \hat{J}_z, \hat{J}_1^2, \hat{J}_2^2 \Rightarrow |j, m_j, j_1, j_2\rangle \quad \text{BASE ACCOPPIATA}$$

COMMUTANO tra di loro

$$\Rightarrow \hat{J}^2 |j, m_j, j_1, j_2\rangle = \hbar^2 j(j+1) |j, m_j, j_1, j_2\rangle$$

$$\hat{J}_z |j, m_j, j_1, j_2\rangle = \hbar m_j |j, m_j, j_1, j_2\rangle$$

$$\hat{J}_1^2 |j, m_j, j_1, j_2\rangle = \hbar^2 j_1(j_1+1) |j, m_j, j_1, j_2\rangle$$

$$\hat{J}_2^2 |j, m_j, j_1, j_2\rangle = \hbar^2 j_2(j_2+1) |j, m_j, j_1, j_2\rangle$$

$$|j, m_j, j_1, j_2\rangle = \sum_{m_1, m_2} \underbrace{C_{m_1, m_2}}_{\text{COEF. di CLEBSCH GORDAN}} |j_1, m_1, j_2, m_2\rangle$$

Se i VETTORI SONO IND. POSSO PASSARE DA UNA BASE ALL'ALTRA

SPIN di DUE ELETTRONI:

$$\begin{cases} S_1 = \frac{1}{2} \\ S_2 = \frac{1}{2} \end{cases}$$

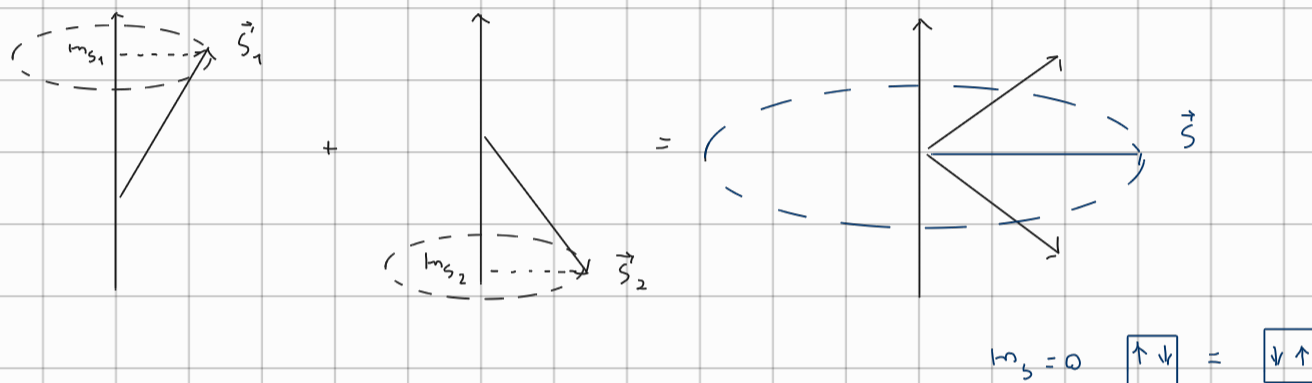
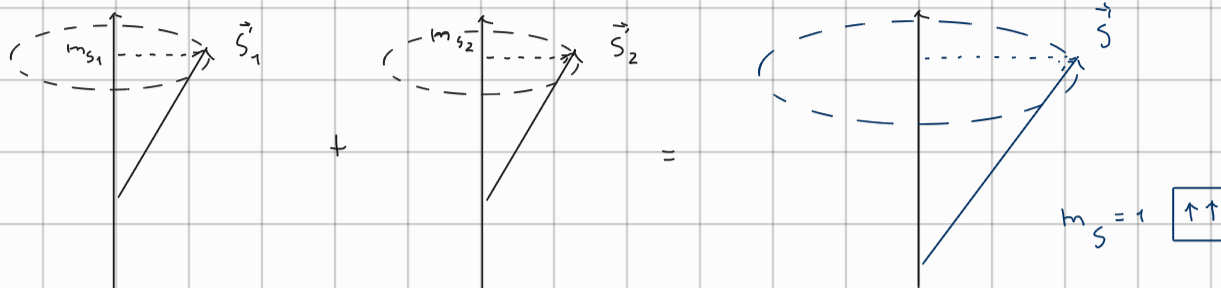
$$S = |S_1 - S_2|, \dots, S_1 + S_2 \Rightarrow S = 0, 1$$

$$\text{Se } S=0 \Rightarrow m_s = 0 \quad \text{SINGOLETTO}$$

$$\text{Se } S=1 \Rightarrow m_s = -1, 0, 1 \quad \text{TRIPLETTO}$$

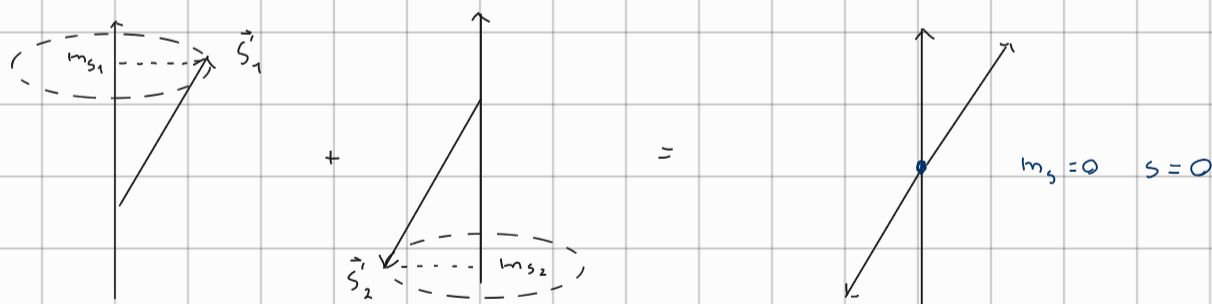
$$m_{s_1} = -\frac{1}{2}, +\frac{1}{2} \quad \Rightarrow \quad m_s = m_{s_1} + m_{s_2} = \begin{cases} 1 & \uparrow \uparrow \\ 0 & \uparrow \downarrow \\ 0 & \downarrow \uparrow \\ -1 & \downarrow \downarrow \end{cases}$$

$$m_{s_2} = -\frac{1}{2}, +\frac{1}{2}$$



UGUALE per $\downarrow + \downarrow = \boxed{\downarrow \downarrow}$

$\Rightarrow$ SINGOLETTO $S=0$, $m_s=0$



MOMENTO ANG. TOTALE:

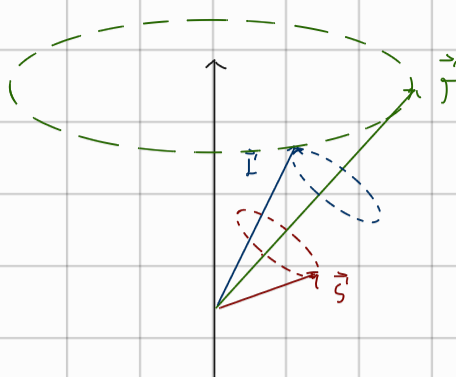
$$\cdot \quad e \quad \begin{array}{l} \vec{L} \\ \vec{S} \end{array} \quad \vec{J} = \vec{L} + \vec{S} \quad j = |l-s|, \dots, l+s = |l-\frac{1}{2}|, \dots, l+\frac{1}{2}$$

POSSO USARE ENTRAMBI BASI ACCOPP. E DISACC.

$\Rightarrow |l, m_l, s, m_s\rangle$ DISACCOPIATA



$\rightarrow |j, m_j, l, s\rangle$ BASE ACCOPPIATA



$\Rightarrow$ Da questa dis. si

capisce perché L_z e S_z

non sono definiti

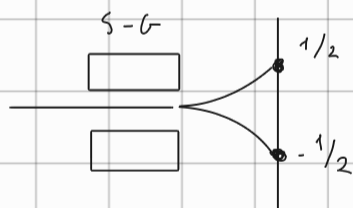
(m_l e m_s non compaiono
nella base)

$$\vec{\mu}_j = -g_j \mu_B \frac{\vec{J}}{\hbar} \quad g_l = 1, g_s = 2 \quad g_j = 1 + \frac{j(j+1) - l(l+1) + s(s+1)}{2j(j+1)}$$

$$(\mu_j)_z = -g_j \mu_B m_j$$

FAITORE DI LANDÉ

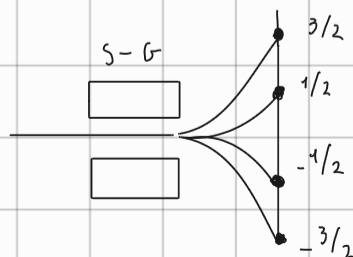
Es.: • H $1s \uparrow$ $l=0, s=\frac{1}{2} \Rightarrow j=\frac{1}{2} \Rightarrow g_j = g_s = 2, m_j = m_s = \pm \frac{1}{2}$



• H $2p \uparrow \downarrow$ $l=1, s=\frac{1}{2} \Rightarrow j=\frac{3}{2}, \frac{1}{2}$

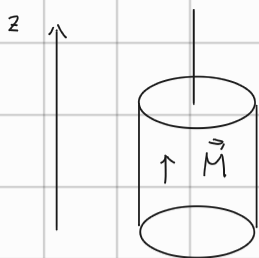
$$j=\frac{3}{2} \Rightarrow m_j = -\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}$$

$$(\mu_j)_z = -g_j \mu_B m_j$$



L'esperimento conferma la teoria

EFFETTO EINSSTEIN-DE HASS



dens. di dip. mag.

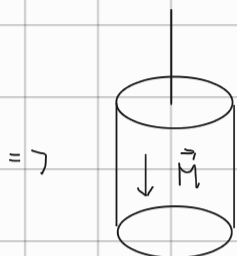
$$\vec{M} = \hbar \vec{M}$$

$$\langle \vec{L} \rangle = 0 \text{ QUENCHING di } \vec{L}$$

$$\vec{M} = \hbar \vec{M}_S \Rightarrow S_z = -\frac{\hbar}{2} \quad (\mu_S \text{ è opposto al campo mag.})$$

$$(\mu_B)_z = -g_S \mu_B m_S$$

$$M = -g_S \mu_B m_S$$



$$S_z = -\frac{\hbar}{2} \Rightarrow S_z = \frac{\hbar}{2} \Rightarrow \Delta M = g_S \mu_B \frac{\Delta S_z}{\hbar} = g_S \mu_B \Delta m_S$$

IL MOM ANG. TOT. SI DEVE CONS. (SIST. ISOLATO)

$$\Delta J_z = \Delta L_z + \Delta S_z = 0$$

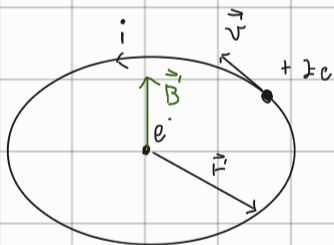
$$\Downarrow L_z = -\Delta S_z = -\frac{\hbar \Delta M}{g_S \mu_B} \vec{n}_z$$

SE CAMBIO LA MAGNETIZ. IL CILINDRO RUOTA

$$\omega = \frac{L_z}{I}$$

INTERAZIONE di SPIN-ORBITA:

solo nel sis. di rif. di e^-



$$\vec{B} = \frac{\mu_0 i}{2r} \hat{n}_L, \quad i = \frac{Ze}{\pi r} = \frac{Ze}{\frac{2\pi r}{v}} = \frac{Zev}{2\pi r}$$

$$|\vec{B}| = \frac{\mu_0 Zev}{4\pi r^2} \frac{mvr}{m_e r} = \frac{\mu_0 Ze}{4\pi \hbar r^3} (mvr) \cdot \frac{\epsilon_0}{\epsilon_0} = \frac{\epsilon_0 \mu_0 Ze}{4\pi \epsilon_0 \hbar r^3} mvr = \frac{Ze}{4\pi \epsilon_0 c^2 \hbar r^3} mvr$$

$$|\vec{B}| = \frac{Ze}{4\pi \epsilon_0 c^2 \hbar r^3} |\vec{L}| \Rightarrow \vec{B} = \frac{Ze}{4\pi \epsilon_0 c^2 \hbar r^3} \vec{L}$$

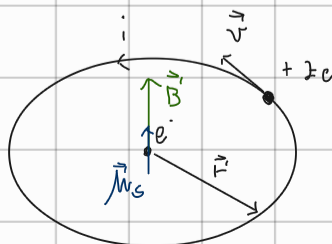
CONSIDERO $\vec{M}_S$:

$$V = -\vec{M}_S \cdot \vec{B} \quad (\text{HO TROVATO QUALCOSA CHE}$$

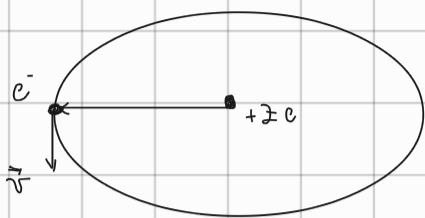
↓
dipende da

dipende da ℓ

ACCOPPIA LO SPIN E LE ORBITA



$$\Rightarrow \Delta E_{so} = - \vec{\mu}_s \cdot \vec{B}$$



Per tornare con il prob. in quiete servono le trasform. di LORENTZ. Se si fanno i conti:

$$\Delta E_{so} = - \frac{1}{2} \vec{\mu}_s \cdot \vec{B}$$

↳ Accensione di Thomas

$$\Delta E_{so} = - \frac{1}{2} \vec{\mu}_s \cdot \vec{B} = \frac{1}{2} g_s \mu_B \frac{Ze}{4\pi\epsilon_0 c^2 m r^3} \vec{L}$$

$$\Delta E_{so} = \frac{Ze^2}{8\pi\epsilon_0 m^2 c^2 r^3} \vec{S} \cdot \vec{L}$$

$$\Rightarrow S \sim L \sim \hbar, \quad r \sim a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m e^2}, \quad E \sim R_{Hy} = \frac{m c^4}{2(4\pi\epsilon_0 \hbar)^2}$$

$$\Delta E_{so} \approx \frac{Ze^2}{8\pi\epsilon_0 m^2 c^2} \cdot \hbar^2 = \frac{Z m^3 e^8 \hbar^2}{8\pi (4\pi)^3 \epsilon_0^4 m^2 c^3 \hbar^6} = \frac{Z m e^8}{2(4\pi)^4 \epsilon_0^4 c^2 \hbar^4} =$$

$$= Z \left[\frac{m c^4}{2(4\pi\epsilon_0 \hbar)^2} \right] \left[\frac{e}{4\pi\epsilon_0 c \hbar} \right]^2 = Z R_{Hy} \alpha^2 \Rightarrow \frac{\Delta E}{E} \approx \alpha^2 \approx 10^{-4}$$

R_{Hy}

$$\alpha \equiv \frac{1}{137}$$

Variaz. piccole ma non trascur.

$$\Delta E_{so} = \frac{1}{2m^2 c^2} \frac{Ze^2}{4\pi\epsilon_0 r^2} \vec{S} \cdot \vec{L}$$

$$\Rightarrow \Delta E_{so} = \frac{1}{2m^2 c^2} \left(\frac{1}{r} \frac{dV}{dr} \right) \vec{S} \cdot \vec{L}$$

$$F_{el} = -|\nabla V| = \frac{dV}{dr}$$

$$\hat{H}_{so} = \frac{Ze^2}{8\pi\epsilon_0 m^2 c^2 r^3} \vec{S} \cdot \vec{L}$$

$$\Rightarrow \hat{H} = -\frac{\hbar^2}{2m} \Delta + \hat{V}(r) + \hat{H}_{so}$$

(NON C'È UNA SOL. ANALITICA DIRETTA)

Lo risolv. con la teor. delle pert. non dipende dal tempo

→ Si utilizza la BASE ACCOPPIATA (se l sono legati) $\Rightarrow |j, m; l, s\rangle$

$$\vec{L} \uparrow \Rightarrow \vec{B} \uparrow ; \vec{W}_s \downarrow \quad \Delta E_{so} = -\vec{W}_s \cdot \vec{B} > \text{POSITIVO (Ho un sum. dell'Eh.)}$$

$$j = l + \frac{1}{2}$$

$$2\Delta E_{so}$$

DOPPIETTI di SPIN ORBITA

$$j = l - \frac{1}{2}$$

$$\vec{L} \uparrow = \vec{B} \uparrow ; \vec{W}_s \uparrow (s \downarrow) \quad \Delta E_{so} \text{ è NEGATIVA (stato di Eh. minore)}$$